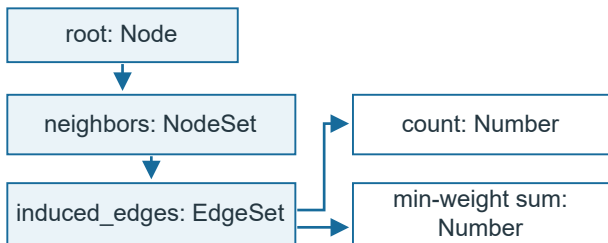
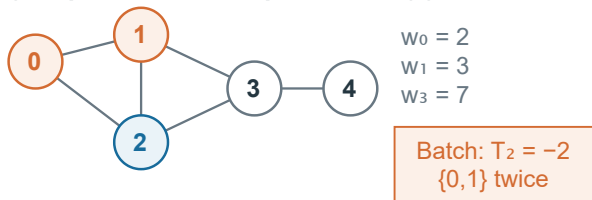


(a) Share a typed expression DAG



Shared nodes; cache per residual state.

(b) Sequential deletion preserves $T(2)$



$T_2: 5 \rightarrow \text{delete } 0: 3 \rightarrow \text{delete } 1: 0$

$T(v) \leftarrow T(v) - \sum \min(w_x, w_y)$
 $y \in N_R(v) \cap N_R(x)$, before deleting x

Charge initialization, queries and updates.

Unrecognized expressions: interpreter.